

Improving Language Understanding by Generative Pre-Training

Alec Radford
OpenAI
alec@openai.com

Karthik Narasimhan
OpenAI
karthikn@openai.com

Tim Salimans
OpenAI
tim@openai.com

Ilya Sutskever
OpenAI
ilyasu@openai.com

Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.¹ Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

¹<https://gluebenchmark.com/leaderboard>

通过生成式预训练提升语言理解能力

Alec RadfordOpenAI
alec@openai.com

Karthik NarasimhanOpenAI
karthikn@openai.com

Tim SalimansOpenAI
tim@openai.com

Ilya SutskeverOpenAI
ilyasu@openai.com

摘要

自然语言理解涵盖了一系列多样化的任务，包括文本蕴含、问答任务、语义相似度评估以及文档分类等。尽管大规模未标注文本语料库十分丰富，但用于学习这些特定任务的标注数据却极为稀缺，这使得基于判别学习的模型难以达到理想的表现。我们证明了，通过对语言模型在多种未标注文本语料库上进行生成式预训练，随后针对每项具体任务进行基于判别学习的微调，便能在这些任务上取得显著提升。与以往的方法不同，我们在微调过程中采用了面向任务的输入转换策略，从而在几乎不改变模型架构的情况下实现有效的知识迁移。我们在各类自然语言理解基准测试中验证了该方法的有效性。我们所设计的通用任务无关模型性能优于那些为每项任务专门设计架构的基于判别学习的模型，在所研究的12项任务中有9项的性能均大幅超越了现有最佳水平。例如，在常识推理（故事填空测试）任务上，我们的方法实现了8.9%的绝对提升；在问答任务（RACE）上提升了5.7%；在文本蕴含（MultiNLI）任务上则提升了1.5%。

1 引言

能够从原始文本中高效地学习知识，对于减少自然语言处理（NLP）领域对监督学习的依赖至关重要。大多数深度学习方法都需要大量的人工标注数据，而这恰恰限制了它们在那些标注资源极为匮乏的领域的应用 [61]。在这样的情况下，那些能够利用未标记数据中的语言信息的模型，便成为了一种极具价值的替代方案，无需耗费大量时间和金钱去收集标注数据。此外，即便有充足的监督数据可用，以无监督的方式学习出优质的特征表示也能显著提升模型性能。迄今为止，最有力的证据便是广泛使用预训练的词嵌入 [10, 39, 42]，以此提升各类NLP任务的性能 [8, 11, 26, 45]。

然而，要利用未标注文本中的词级信息之外的更多内容来提升性能，主要面临两大挑战。首先，目前尚不清楚何种优化目标最能有效用于学习那些便于迁移的文本表示。现有研究探讨了多种目标，比如语言建模、机器翻译以及篇章连贯性等，而这些方法在不同任务上的表现也各有优劣。其次，关于如何最有效地将这些学到的表示迁移到目标任务上，目前仍缺乏共识。现有的技术通常结合对模型架构进行针对特定任务的调整、采用复杂的学习机制，以及增设辅助学习目标等多种手段。正是这些不确定性，使得为语言处理开发有效的半监督学习方法变得十分困难。

¹<https://gluebenchmark.com/leaderboard>

In this paper, we explore a semi-supervised approach for language understanding tasks using a combination of unsupervised pre-training and supervised fine-tuning. Our goal is to learn a universal representation that transfers with little adaptation to a wide range of tasks. We assume access to a large corpus of unlabeled text and several datasets with manually annotated training examples (target tasks). Our setup does not require these target tasks to be in the same domain as the unlabeled corpus. We employ a two-stage training procedure. First, we use a language modeling objective on the unlabeled data to learn the initial parameters of a neural network model. Subsequently, we adapt these parameters to a target task using the corresponding supervised objective.

For our model architecture, we use the *Transformer* [62], which has been shown to perform strongly on various tasks such as machine translation [62], document generation [34], and syntactic parsing [29]. This model choice provides us with a more structured memory for handling long-term dependencies in text, compared to alternatives like recurrent networks, resulting in robust transfer performance across diverse tasks. During transfer, we utilize task-specific input adaptations derived from traversal-style approaches [52], which process structured text input as a single contiguous sequence of tokens. As we demonstrate in our experiments, these adaptations enable us to fine-tune effectively with minimal changes to the architecture of the pre-trained model.

We evaluate our approach on four types of language understanding tasks – natural language inference, question answering, semantic similarity, and text classification. Our general task-agnostic model outperforms discriminatively trained models that employ architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test) [40], 5.7% on question answering (RACE) [30], 1.5% on textual entailment (MultiNLI) [66] and 5.5% on the recently introduced GLUE multi-task benchmark [64]. We also analyzed zero-shot behaviors of the pre-trained model on four different settings and demonstrate that it acquires useful linguistic knowledge for downstream tasks.

2 Related Work

Semi-supervised learning for NLP Our work broadly falls under the category of semi-supervised learning for natural language. This paradigm has attracted significant interest, with applications to tasks like sequence labeling [24, 33, 57] or text classification [41, 70]. The earliest approaches used unlabeled data to compute word-level or phrase-level statistics, which were then used as features in a supervised model [33]. Over the last few years, researchers have demonstrated the benefits of using word embeddings [11, 39, 42], which are trained on unlabeled corpora, to improve performance on a variety of tasks [8, 11, 26, 45]. These approaches, however, mainly transfer word-level information, whereas we aim to capture higher-level semantics.

Recent approaches have investigated learning and utilizing more than word-level semantics from unlabeled data. Phrase-level or sentence-level embeddings, which can be trained using an unlabeled corpus, have been used to encode text into suitable vector representations for various target tasks [28, 32, 1, 36, 22, 12, 56, 31].

Unsupervised pre-training Unsupervised pre-training is a special case of semi-supervised learning where the goal is to find a good initialization point instead of modifying the supervised learning objective. Early works explored the use of the technique in image classification [20, 49, 63] and regression tasks [3]. Subsequent research [15] demonstrated that pre-training acts as a regularization scheme, enabling better generalization in deep neural networks. In recent work, the method has been used to help train deep neural networks on various tasks like image classification [69], speech recognition [68], entity disambiguation [17] and machine translation [48].

The closest line of work to ours involves pre-training a neural network using a language modeling objective and then fine-tuning it on a target task with supervision. Dai et al. [13] and Howard and Ruder [21] follow this method to improve text classification. However, although the pre-training phase helps capture some linguistic information, their usage of LSTM models restricts their prediction ability to a short range. In contrast, our choice of transformer networks allows us to capture longer-range linguistic structure, as demonstrated in our experiments. Further, we also demonstrate the effectiveness of our model on a wider range of tasks including natural language inference, paraphrase detection and story completion. Other approaches [43, 44, 38] use hidden representations from a

在本文中，我们探索了一种结合无监督预训练与有监督微调的半监督方法，用以解决语言理解任务中的相关问题。我们的目标是学习一种通用表示，使其能够在几乎无需额外调整的情况下应用于各种不同的任务。我们假设能够获取大规模的未标注文本语料库，以及若干包含人工标注训练样本（即目标任务）的数据集。我们的方法并不要求这些目标任务与未标注语料库属于同一领域。我们采用了两阶段训练流程：首先在未标注数据上运用语言建模目标来学习神经网络模型的初始参数；随后再利用相应的有监督目标，将这些参数适配到具体的目标任务中。

在模型架构方面，我们采用了*Transformer* [62] 架构，该架构已在机器翻译 [62]、文档生成 [34]、以及句法分析 [29]等多种任务中展现出出色的性能。与循环网络等替代方案相比，这种模型选择能为处理文本中的长期依赖关系提供更为结构化的记忆机制，从而在各类任务中实现更为稳健的迁移学习效果。在迁移学习过程中，我们运用了基于遍历式方法 [52]，开发的任务特定输入适配技术，这类技术能够将结构化文本输入视为连续的标记序列进行处理。实验结果表明，通过这些适配技术，我们只需对预训练模型的架构进行极小的修改，即可实现高效的微调。

我们在四类语言理解任务上对所提出的方法进行了评估，这四类任务分别是自然语言推理、问答任务、语义相似度计算以及文本分类。我们的通用任务无关模型在性能上显著优于那些为每项任务专门设计架构的基于判别学习的模型，在所研究的12项任务中有9项的性能均超过了当前最优水平。例如，它在常识推理（故事填空测试） [40]、方面的提升幅度达到了8.9%，在问答任务（RACE） [30]、1方面的提升率为5.7%，在文本蕴含（MultiNLI） [66] 方面的提升率为0.5%，而在最新推出的GLUE多任务基准测试 [64]中的提升率则为5.5%。此外，我们还分析了该预训练模型在四种不同设置下的零样本行为，证实它确实能够为后续任务获取有用的语言知识。

2 相关工作

自然语言处理的半监督学习 我们的研究主要属于自然语言领域中的半监督学习范畴。这一研究范式备受关注，已被应用于序列标注 [24, 33, 57] 或文本分类 [41, 70]等目标任务中。早期的方法会利用未标记数据来计算词级或短语级的统计特征，这些特征随后会被用作监督学习模型中的输入 [33]。近几年来，研究人员通过使用在未标记语料库上训练得到的词嵌入 [11, 39, 42]，证明了其在提升各类任务性能方面的优势 [8, 11, 26, 45]。不过，这些方法主要传递的是词级信息，而我们的目标是捕捉更高级别的语义含义。

近期的研究开始探索如何从未标记数据中学习并利用超出词级语义的内容。借助可在未标记语料库上训练得到的短语级或句子级嵌入向量，人们能够将文本编码为适合多种目标任务使用的向量表示 [28,32, 1, 36, 22, 12, 56, 31]。

无监督预训练 无监督预训练属于半监督学习的一种特殊形式，其目标并非修改有监督学习目标，而是寻找一个合适的初始化点。早期的研究探索了将该技术应用于图像分类 [20, 49, 63] 以及回归任务 [3]中的可能性。后续的研究 [15] 表明，预训练实际上是一种正则化机制，能够帮助深度神经网络实现更好的泛化能力。在近期的研究中，这种方法已被用于协助训练处理各种任务的深度神经网络，这些任务包括图像分类 [69]、语音识别 [68]、实体消歧 [17] 以及机器翻译 [48]。

与我们的工作最为接近的研究方向是：先以语言建模目标对神经网络进行预训练，随后在带有监督信号的目标任务上对其进行微调。Dai等人 [13] 以及Howard和Ruder [21] 也采用了这种方法来提升文本分类的性能。不过，尽管预训练阶段有助于捕捉部分语言信息，但他们所使用的LSTM模型使得模型的预测能力仅限于较短的范围。相比之下，我们选择的变换器网络能够捕捉更长范围的语言结构，这一点已在我们的实验中得到验证。此外，我们还证明了该模型在包括自然语言推理、意译检测和故事补全在内的更多任务上的有效性。其他一些方法 [43, 44, 38] 则会在针对目标任务训练有监督模型时，使用来自预训练过的语言模型或机器翻译模型的隐藏表示作为辅助特征。这类方法为每个单独的目标任务都需要大量新的参数，而我们在模型迁移过程中仅需对模型架构进行极少的修改。

pre-trained language or machine translation model as auxiliary features while training a supervised model on the target task. This involves a substantial amount of new parameters for each separate target task, whereas we require minimal changes to our model architecture during transfer.

Auxiliary training objectives Adding auxiliary unsupervised training objectives is an alternative form of semi-supervised learning. Early work by Collobert and Weston [10] used a wide variety of auxiliary NLP tasks such as POS tagging, chunking, named entity recognition, and language modeling to improve semantic role labeling. More recently, Rei [50] added an auxiliary language modeling objective to their target task objective and demonstrated performance gains on sequence labeling tasks. Our experiments also use an auxiliary objective, but as we show, unsupervised pre-training already learns several linguistic aspects relevant to target tasks.

3 Framework

Our training procedure consists of two stages. The first stage is learning a high-capacity language model on a large corpus of text. This is followed by a fine-tuning stage, where we adapt the model to a discriminative task with labeled data.

3.1 Unsupervised pre-training

Given an unsupervised corpus of tokens $\mathcal{U} = \{u_1, \dots, u_n\}$, we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

where k is the size of the context window, and the conditional probability P is modeled using a neural network with parameters Θ . These parameters are trained using stochastic gradient descent [51].

In our experiments, we use a multi-layer *Transformer decoder* [34] for the language model, which is a variant of the transformer [62]. This model applies a multi-headed self-attention operation over the input context tokens followed by position-wise feedforward layers to produce an output distribution over target tokens:

$$\begin{aligned} h_0 &= UW_e + W_p \\ h_l &= \text{transformer_block}(h_{l-1}) \forall i \in [1, n] \\ P(u) &= \text{softmax}(h_n W_e^T) \end{aligned} \quad (2)$$

where $U = (u_{-k}, \dots, u_{-1})$ is the context vector of tokens, n is the number of layers, W_e is the token embedding matrix, and W_p is the position embedding matrix.

3.2 Supervised fine-tuning

After training the model with the objective in Eq. 1, we adapt the parameters to the supervised target task. We assume a labeled dataset $\mathcal{C}$, where each instance consists of a sequence of input tokens, $x^1, \dots, x^m$, along with a label y . The inputs are passed through our pre-trained model to obtain the final transformer block’s activation h_l^m , which is then fed into an added linear output layer with parameters W_y to predict y :

$$P(y | x^1, \dots, x^m) = \text{softmax}(h_l^m W_y). \quad (3)$$

This gives us the following objective to maximize:

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y | x^1, \dots, x^m). \quad (4)$$

We additionally found that including language modeling as an auxiliary objective to the fine-tuning helped learning by (a) improving generalization of the supervised model, and (b) accelerating convergence. This is in line with prior work [50, 43], who also observed improved performance with such an auxiliary objective. Specifically, we optimize the following objective (with weight λ):

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \quad (5)$$

Overall, the only extra parameters we require during fine-tuning are W_y , and embeddings for delimiter tokens (described below in Section 3.3).

在针对目标任务训练监督模型时，会将预训练的语言模型或机器翻译模型作为辅助特征使用。这种方式意味着每个独立的任务都需要大量新的参数，而在进行迁移学习时，我们的模型架构却仅需极小的调整即可。

辅助训练目标 引入辅助性的无监督训练目标是一种半监督学习的变体形式。Collobert和Weston在早期的研究中 [10] 就运用了多种辅助性的自然语言处理任务，比如词性标注、分词、命名实体识别以及语言模型构建，以此来提升语义角色标注的准确性。近期，Rei [50] 在其目标任务的优化目标中加入了额外的语言模型构建目标，并在序列标注任务上取得了性能提升。我们的实验同样采用了辅助目标，但正如我们所展示的，无监督预训练本身就已经能够学到与目标任务相关的诸多语言特征。

3 框架

我们的训练流程分为两个阶段。第一阶段是在大规模文本语料库上训练出具有高容量的语言模型；随后进入微调阶段，在此阶段我们会利用标注数据将该模型适配到特定的判别性任务中。

3.1 无监督预训练

给定一个由标记组成的无监督语料库 $\mathcal{U} = \{u_1, \dots, u_n\}$ 后，我们会使用标准的语言建模目标来最大化以下概率值：

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta) \quad (1)$$

其中 k 表示上下文窗口的大小，而条件概率 P 则通过带有参数 Θ 的神经网络来建模。这些参数是借助随机梯度下降 [51]方法进行训练的。

在我们的实验中，语言模型采用了多层*Transformer*解码器 [34]，它其实是Transformer架构 [62]的一种变体。该模型首先对输入的上下文标记应用多头自注意力机制，随后通过逐位置的前馈层处理，从而生成目标标记的输出分布。

$$\begin{aligned} h_0 &= UW_e + W_p \\ h_l &= \text{transformer_block}(h_{l-1}) \forall i \in [1, n] \\ P(u) &= \text{softmax}(h_n W_e^T) \end{aligned} \quad (2)$$

此处 $U = (u_{-k}, \dots, u_{-1})$ 代表标记的上下文向量， n 为层数， W_e 是标记嵌入矩阵，而 W_p 则是位置嵌入矩阵。

3.2 有监督微调

在根据公式1中的目标函数完成模型训练后，我们会将这些参数适配到有监督的目标任务中。我们假设存在一个标注数据集 $\mathcal{C}$ ，其中的每个样本都包含一串输入标记 $x^1, \dots, x^m$ 以及对应的标签 y 。这些输入会被送入我们的预训练模型，以获得最后一个Transformer模块的激活值 h_l^m ，之后该激活值会被传递到一个带有参数 W_y 的线性输出层中，用于预测 y 。

$$P(y | x^1, \dots, x^m) = \text{softmax}(h_l^m W_y). \quad (3)$$

这为我们确立了以下需要最大化的目标：

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y | x^1, \dots, x^m). \quad (4)$$

我们进一步发现，在微调过程中将语言模型作为辅助目标，有助于提升学习效果——具体而言，既能提高受监督模型的泛化能力，又能加速收敛速度。这一结论与之前的研究 [50, 43], 的结果一致，那些研究也指出采用此类辅助目标能够提升性能。具体来说，我们会对以下目标进行优化（权重为 λ ）：

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \quad (5)$$

总体而言，在微调过程中唯一需要的额外参数就是 W_y ，以及用于分隔符标记的嵌入向量（具体内容见第3.3节）。

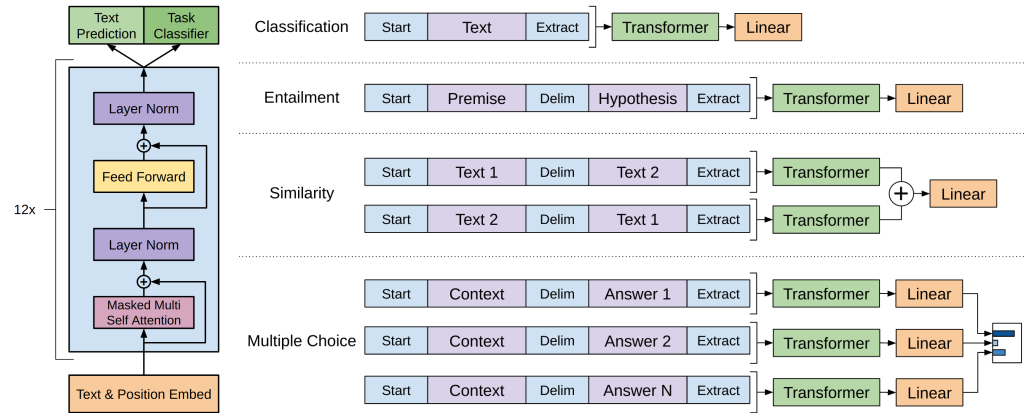


Figure 1: **(left)** Transformer architecture and training objectives used in this work. **(right)** Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

3.3 Task-specific input transformations

For some tasks, like text classification, we can directly fine-tune our model as described above. Certain other tasks, like question answering or textual entailment, have structured inputs such as ordered sentence pairs, or triplets of document, question, and answers. Since our pre-trained model was trained on contiguous sequences of text, we require some modifications to apply it to these tasks. Previous work proposed learning task specific architectures on top of transferred representations [44]. Such an approach re-introduces a significant amount of task-specific customization and does not use transfer learning for these additional architectural components. Instead, we use a traversal-style approach [52], where we convert structured inputs into an ordered sequence that our pre-trained model can process. These input transformations allow us to avoid making extensive changes to the architecture across tasks. We provide a brief description of these input transformations below and Figure 1 provides a visual illustration. All transformations include adding randomly initialized start and end tokens ($\langle s \rangle$, $\langle e \rangle$).

Textual entailment For entailment tasks, we concatenate the premise p and hypothesis h token sequences, with a delimiter token ($\$$) in between.

Similarity For similarity tasks, there is no inherent ordering of the two sentences being compared. To reflect this, we modify the input sequence to contain both possible sentence orderings (with a delimiter in between) and process each independently to produce two sequence representations h_l^m which are added element-wise before being fed into the linear output layer.

Question Answering and Commonsense Reasoning For these tasks, we are given a context document z , a question q , and a set of possible answers $\{a_k\}$. We concatenate the document context and question with each possible answer, adding a delimiter token in between to get $[z; q; \$; a_k]$. Each of these sequences are processed independently with our model and then normalized via a softmax layer to produce an output distribution over possible answers.

4 Experiments

4.1 Setup

Unsupervised pre-training We use the BooksCorpus dataset [71] for training the language model. It contains over 7,000 unique unpublished books from a variety of genres including Adventure, Fantasy, and Romance. Crucially, it contains long stretches of contiguous text, which allows the generative model to learn to condition on long-range information. An alternative dataset, the 1B Word Benchmark, which is used by a similar approach, ELMo [44], is approximately the same size

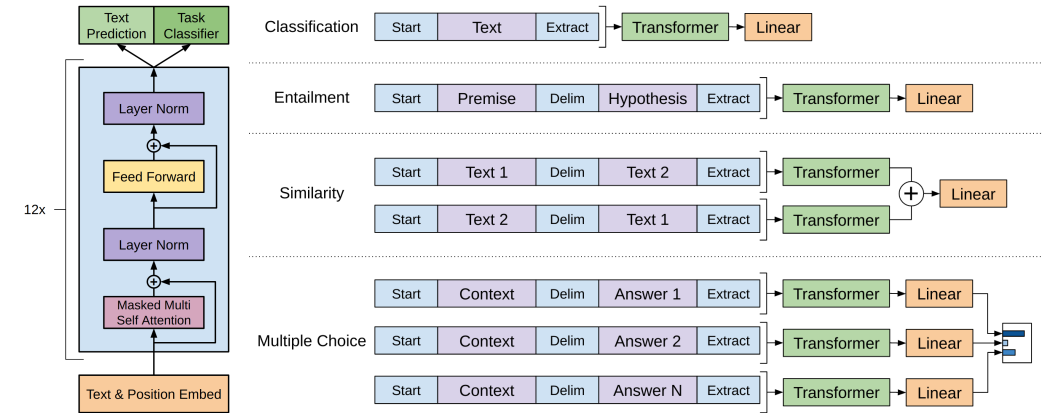


图1: (左) 本研究所采用的Transformer架构及训练目标。(右) 针对不同任务进行微调时的输入转换方式。我们会将所有结构化输入转换为词元序列，以便由预训练模型进行处理，之后再通过线性+softmax层进行处理。

3.3 任务特定的输入转换

对于文本分类这类任务，我们可以按照上述方法直接对模型进行微调。而问答任务或文本蕴含任务则属于需要结构化输入的任务，这类输入表现为有序句子对，或是文档、问题与答案三元组的形式。由于我们的预训练模型是基于连续的文本序列训练的，因此若要将其应用于这些任务，就必须对其进行一些修改。以往的研究提出在迁移学习得到的表示基础上进一步学习特定于任务的架构 [44]，但这种做法会引入大量针对特定任务的定制内容，且并未利用迁移学习来处理这些额外的架构组件。相反，我们采用了遍历式方法 [52]，通过将该类结构化输入转换为预训练模型能够处理的有序序列，从而避免在不同任务之间对架构进行大规模调整。下文将简要介绍这些输入转换方法，图1则提供了相应的可视化示例。所有转换步骤均包括添加随机初始化的起始词元和结束词元($\langle s \rangle$, $\langle e \rangle$)。

文本蕴含 在处理文本蕴含任务时，我们会将前提 p 与假设 h 的词元序列拼接起来，并在二者之间添加分隔符词元($\$$)。

相似度计算 对于相似度计算任务，被比较的两句话本身并不存在固定的顺序。为体现这一特点，我们会修改输入序列，使其包含两种可能的句子顺序（并在两者之间添加分隔符），随后分别对这两种顺序进行处理，从而生成两个 h_l^m 类型的序列表示，最后将这些表示逐元素相加后再输入到线性输出层中。

问答任务与常识推理 对于这类任务，系统会收到上下文文档 z 、一个问题 q 以及一组可能的答案 $\{a_k\}$ 。我们会将文档内容、问题以及每个可能的答案拼接在一起，并在它们之间添加分隔符词元，最终得到 $[z; q; \$; a_k]$ 这样的序列。这些序列会分别通过我们的模型进行处理，之后再通过softmax层进行归一化，从而生成关于各可能答案的输出概率分布。

4 实验

4.1 设置

无监督预训练 我们使用BooksCorpus数据集 [71] 来训练语言模型。该数据集包含了7,000多本风格各异的未出版书籍，涵盖冒险、奇幻和言情等多种类型。尤为重要的是，其中含有大量连续的文本片段，这有助于生成模型学会依据长距离信息进行推理。另一种类似方法所使用的10亿词基准数据集 [44]，——即ELMo模型所依赖的数据集——规模与之大致相当，但其内容是在句子层面被打乱顺序的，从而导致长距离结构被破坏。而我们的语言模型在该语料库上的词级困惑度仅仅为18.4，数值极低。

Table 1: A list of the different tasks and datasets used in our experiments.

Task	Datasets
Natural language inference	SNLI [5], MultiNLI [66], Question NLI [64], RTE [4], SciTail [25]
Question Answering	RACE [30], Story Cloze [40]
Sentence similarity	MSR Paraphrase Corpus [14], Quora Question Pairs [9], STS Benchmark [6]
Classification	Stanford Sentiment Treebank-2 [54], CoLA [65]

but is shuffled at a sentence level - destroying long-range structure. Our language model achieves a very low token level perplexity of 18.4 on this corpus.

Model specifications Our model largely follows the original transformer work [62]. We trained a 12-layer decoder-only transformer with masked self-attention heads (768 dimensional states and 12 attention heads). For the position-wise feed-forward networks, we used 3072 dimensional inner states. We used the Adam optimization scheme [27] with a max learning rate of 2.5e-4. The learning rate was increased linearly from zero over the first 2000 updates and annealed to 0 using a cosine schedule. We train for 100 epochs on minibatches of 64 randomly sampled, contiguous sequences of 512 tokens. Since layernorm [2] is used extensively throughout the model, a simple weight initialization of $N(0, 0.02)$ was sufficient. We used a bytetrain encoding (BPE) vocabulary with 40,000 merges [53] and residual, embedding, and attention dropouts with a rate of 0.1 for regularization. We also employed a modified version of L2 regularization proposed in [37], with $w = 0.01$ on all non bias or gain weights. For the activation function, we used the Gaussian Error Linear Unit (GELU) [18]. We used learned position embeddings instead of the sinusoidal version proposed in the original work. We use the *ftfy* library² to clean the raw text in BooksCorpus, standardize some punctuation and whitespace, and use the *spaCy* tokenizer.³

Fine-tuning details Unless specified, we reuse the hyperparameter settings from unsupervised pre-training. We add dropout to the classifier with a rate of 0.1. For most tasks, we use a learning rate of 6.25e-5 and a batchsize of 32. Our model finetunes quickly and 3 epochs of training was sufficient for most cases. We use a linear learning rate decay schedule with warmup over 0.2% of training. λ was set to 0.5.

4.2 Supervised fine-tuning

We perform experiments on a variety of supervised tasks including natural language inference, question answering, semantic similarity, and text classification. Some of these tasks are available as part of the recently released GLUE multi-task benchmark [64], which we make use of. Figure 1 provides an overview of all the tasks and datasets.

Natural Language Inference The task of natural language inference (NLI), also known as recognizing textual entailment, involves reading a pair of sentences and judging the relationship between them from one of `entailment`, `contradiction` or `neutral`. Although there has been a lot of recent interest [58, 35, 44], the task remains challenging due to the presence of a wide variety of phenomena like lexical entailment, coreference, and lexical and syntactic ambiguity. We evaluate on five datasets with diverse sources, including image captions (SNLI), transcribed speech, popular fiction, and government reports (MNLI), Wikipedia articles (QNLI), science exams (SciTail) or news articles (RTE).

Table 2 details various results on the different NLI tasks for our model and previous state-of-the-art approaches. Our method significantly outperforms the baselines on four of the five datasets, achieving absolute improvements of upto 1.5% on MNLI, 5% on SciTail, 5.8% on QNLI and 0.6% on SNLI over the previous best results. This demonstrates our model’s ability to better reason over multiple sentences, and handle aspects of linguistic ambiguity. On RTE, one of the smaller datasets we evaluate on (2490 examples), we achieve an accuracy of 56%, which is below the 61.7% reported by a multi-task biLSTM model. Given the strong performance of our approach on larger NLI datasets, it is likely our model will benefit from multi-task training as well but we have not explored this currently.

²<https://ftfy.readthedocs.io/en/latest/>

³<https://spacy.io/>

表1：我们实验中使用的各类任务与数据集列表。

Task	数据集
自然语言推理	SNLI数据集 [5],MultiNLI测验 [66],Question NLI数据集 [64], RTE数据集 [4], SciTail数据集 [25]
问答任务	RACE测验 [30],Story Cloze任务 [40]
句子相似度	MSR释义语料库 [14]、Quora问题对数据集 [9], 以及STS基准测试 [6]
分类任务	斯坦福情感树库-2 [54]、CoLA任务 [65]

我们的语言模型在该语料库上的词级困惑度仅为18.4，数值十分低。

模型规格 我们的模型在很大程度上沿用了最初的Transformer架构研究 [62]。我们训练了一个仅包含解码器的12层Transformer模型，该模型配备了带掩码的自注意力模块（状态维度为768，注意力头数为12）。对于位置相关的前馈网络，我们采用了维度为3072的内部状态。模型采用Adam优化算法 [27]，最大学习率为2.5e-4。在学习率方面，它在前2000次迭代中会从0开始线性上升，之后再通过余弦衰减策略逐渐降至0。训练过程中，我们会使用随机抽取的、长度为512且连续的序列构成的小批量数据，每批64个样本，共进行100轮训练。由于整个模型中广泛使用了层归一化 [2]，因此采用简单的权重初始化方式 $N(0, 0.02)$ 即可满足需求。此外，我们还使用了经过40,000次合并操作的字节对编码（BPE）词汇表 [53]，并为残差连接、嵌入层以及注意力层分别设置了0.1的丢弃层比例以实现正则化处理。同时，我们采用了在 [37] 中提出的L2正则化方法的改进版本，对所有非偏置项和增益项应用 $w = 0.01$ 。在激活函数方面，我们选择了高斯误差线性单元（GELU） [18]。与原始研究中的正弦函数版本不同，我们使用了通过学习得到的位置嵌入。最后，我们借助*ftfy* 库² 来清洗BooksCorpus中的原始文本，统一标点符号和空白字符的处理方式，并使用 *spaCy* 分词器³。

微调细节 除非另有说明，否则我们会沿用无监督预训练时的超参数设置。我们会在分类器中加入丢弃层，其丢弃率为0.1。在处理大多数任务时，所采用的学习率为6.25e-5，批量大小为32。该模型训练速度较快，对于多数情况而言，3轮训练就已足够。我们采用了带有预热阶段的线性学习率衰减调度，预热比例占训练总量的0.2%；同时 λ 的值被设定为0.5。

4.2 有监督微调

我们在多种有监督任务上开展了实验，这些任务包括自然语言推理、问答任务、语义相似度计算以及文本分类。其中部分任务包含在近期发布的GLUE多任务基准测试 [64]中，我们也利用了这一基准测试。图1展示了所有任务及对应数据集的概览。

自然语言推理 自然语言推理（NLI），也被称为文本蕴含识别任务，其核心是读取一对句子，然后判断它们之间的关系属于蕴含、矛盾还是中性之一。尽管近期对此类任务的研究兴趣颇高 [58, 35, 44]，但由于存在词汇蕴含关系、共指现象以及词汇与句法歧义等多种复杂现象，该任务依然具有较高的难度。我们的评估基于五个来源各异的数据集，这些数据集包括图片说明文字（SNLI）、转录语音文本、通俗小说以及政府报告（MNLI）、维基百科文章（QNLI）、科学考试题（SciTail）和新闻文章（RTE）。

表2详细列出了我们的模型与此前当前最先进方法在各类自然语言推理任务上的表现结果。在五个数据集四个上，我们的方法均显著优于基线方法，其准确率相比之前的最佳成绩分别提升了1.5%、5%、5.8%和0.6%。这一优异表现充分体现了该模型更强的多句推理能力，以及处理语言歧义问题的能力。在RTE数据集上——这是我们测试的规模较小的数据集之一，仅包含2490条样本——我们的模型取得了56%的准确率，这一数值低于多任务biLSTM模型所报告的61.7%。鉴于我们的方法在规模更大的自然语言推理数据集上表现出色，未来通过多任务训练进一步提升性能很有可能，但目前我们尚未对此展开研究。

²<https://ftfy.readthedocs.io/en/latest/>

<https://spacy.io/>

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

Method	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5x)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5x)	80.2	79.0	<u>89.3</u>	-	-	-
Stochastic Answer Network [35] (3x)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
Multi-task BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
Finetuned Transformer LM (ours)	82.1	81.4	89.9	88.3	88.1	56.0

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

Method	Story Cloze	RACE-m	RACE-h	RACE
val-LS-skip [55]	76.5	-	-	-
Hidden Coherence Model [7]	<u>77.6</u>	-	-	-
Dynamic Fusion Net [67] (9x)	-	55.6	49.4	51.2
BiAttention MRU [59] (9x)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
Finetuned Transformer LM (ours)	86.5	62.9	57.4	59.0

Question answering and commonsense reasoning Another task that requires aspects of single and multi-sentence reasoning is question answering. We use the recently released RACE dataset [30], consisting of English passages with associated questions from middle and high school exams. This corpus has been shown to contain more reasoning type questions that other datasets like CNN [19] or SQuAD [47], providing the perfect evaluation for our model which is trained to handle long-range contexts. In addition, we evaluate on the Story Cloze Test [40], which involves selecting the correct ending to multi-sentence stories from two options. On these tasks, our model again outperforms the previous best results by significant margins - up to 8.9% on Story Cloze, and 5.7% overall on RACE. This demonstrates the ability of our model to handle long-range contexts effectively.

Semantic Similarity Semantic similarity (or paraphrase detection) tasks involve predicting whether two sentences are semantically equivalent or not. The challenges lie in recognizing rephrasing of concepts, understanding negation, and handling syntactic ambiguity. We use three datasets for this task – the Microsoft Paraphrase corpus (MRPC) [14] (collected from news sources), the Quora Question Pairs (QQP) dataset [9], and the Semantic Textual Similarity benchmark (STS-B) [6]. We obtain state-of-the-art results on two of the three semantic similarity tasks (Table 4) with a 1 point absolute gain on STS-B. The performance delta on QQP is significant, with a 4.2% absolute improvement over Single-task BiLSTM + ELMo + Attn.

Classification Finally, we also evaluate on two different text classification tasks. The Corpus of Linguistic Acceptability (CoLA) [65] contains expert judgements on whether a sentence is grammatical or not, and tests the innate linguistic bias of trained models. The Stanford Sentiment Treebank (SST-2) [54], on the other hand, is a standard binary classification task. Our model obtains an score of 45.4 on CoLA, which is an especially big jump over the previous best result of 35.0, showcasing the innate linguistic bias learned by our model. The model also achieves 91.3% accuracy on SST-2, which is competitive with the state-of-the-art results. We also achieve an overall score of 72.8 on the GLUE benchmark, which is significantly better than the previous best of 68.9.

表2：自然语言推理任务上的实验结果，对比了我们的模型与当前最先进的算法。5x表示由5个模型组成的集成模型；所有数据集均以准确率作为评估指标。

方法	MNLI-m	MNLI-mm	SNLI	SciTail	QNLI	RTE
ESIM + ELMo [44] (5倍)	-	-	<u>89.3</u>	-	-	-
CAFE [58] (5倍)	80.2	79.0	<u>89.3</u>	-	-	-
随机答案网络 [35](3倍)	<u>80.6</u>	<u>80.1</u>	-	-	-	-
CAFE [58]	78.7	77.9	88.5	<u>83.3</u>		
GenSen [64]	71.4	71.3	-	-	<u>82.3</u>	59.2
多任务BiLSTM + Attn [64]	72.2	72.1	-	-	82.1	61.7
微调后的Transformer语言模型（我们团队开发的版本）	82.1	81.4	89.9	88.3	88.1	56.0

表3：在问答任务与常识推理任务上的测试结果，对比了我们的模型与当前最先进的算法。其中9x表示由9个模型构成的集成模型。

方法	Story Cloze任务	RACE-m方法	RACE-h方法	RACE
val-LS-skip [55]	76.5	-	-	-
隐藏一致性模型 [7]	<u>77.6</u>	-	-	-
动态融合网络 [67] (9倍规模)	-	55.6	49.4	51.2
双注意力MRU机制 [59] (9倍规模)	-	<u>60.2</u>	<u>50.3</u>	<u>53.3</u>
微调后的Transformer语言模型（我们团队开发的版本）	86.5	62.9	57.4	59.0

问答任务与常识推理 另一类需要运用单句及多句推理能力的任务是问答任务。我们采用了最新发布的RACE数据集 [30],，该数据集包含来自中小学考试中的英文篇章及其对应题目。与其他如CNN [19] 或SQuAD [47], 这类数据集相比，该语料库中含有更多推理类题目，非常适合用来评估我们这款专为处理长距离上下文而设计的模型。此外，我们还对Story Cloze Test [40], 进行了测试，这项任务要求从两个选项中选出多句故事的正确结局。在这些任务上，我们的模型同样以较大的优势超越了此前的最佳成绩，在Story Cloze Test中的提升幅度高达8.9%，在RACE测验中的整体提升幅度则为5.7%。这些结果进一步证明了我们的模型具备有效处理长距离上下文的能力。

语义相似度 语义相似度（亦即意译检测）任务的核心目标是判断两句话在语义上是否等价。这类任务面临的挑战包括识别概念的不同表述方式、理解否定语义以及处理句法歧义。为此我们采用了三个数据集——来自新闻来源的微软释义语料库（MRPC） [14]、Quora问题对数据集（QQP） [9],，以及语义文本相似性基准测试（STS-B） [6]。在三项语义相似度任务中，我们有两项取得了当前最先进的性能（见表4），在STS-B任务上的绝对得分提升了1分；而在QQP任务上的表现提升尤为显著，相较于单任务BiLSTM + ELMo +Attn模型，其精度提升了4.2%。

分类最后，我们还在两项不同的文本分类任务上进行了评估。语言可接受性语料库（CoLA） [65] 中包含专家对句子是否合乎语法的判断，用于检测训练有素的模型所具有的固有语言偏见。而斯坦福情感树库（SST-2） [54], 则属于标准的二分类任务。在我们的模型中，CoLA任务的得分达到了45.4分，这一数值较之前的最佳成绩35.0分有了显著提升，充分体现了该模型所学到的固有语言偏见。同时，该模型在SST-2任务上的准确率达到了91.3%，其表现与当前最先进的成果相当。此外，该模型在GLUE基准测试中的总得分也为72.8分，远超此前的最佳纪录68.9分。

Table 4: Semantic similarity and classification results, comparing our model with current state-of-the-art methods. All task evaluations in this table were done using the GLUE benchmark. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

Method	Classification		Semantic Similarity			GLUE
	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	
Sparse byte mLSTM [16]	-	93.2	-	-	-	-
TF-KLD [23]	-	-	86.0	-	-	-
ECNU (mixed ensemble) [60]	-	-	-	81.0	-	-
Single-task BiLSTM + ELMo + Attn [64]	35.0	90.2	80.2	55.5	66.1	64.8
Multi-task BiLSTM + ELMo + Attn [64]	18.9	91.6	83.5	72.8	63.3	68.9
Finetuned Transformer LM (ours)	45.4	91.3	82.3	82.0	70.3	72.8

Overall, our approach achieves new state-of-the-art results in 9 out of the 12 datasets we evaluate on, outperforming ensembles in many cases. Our results also indicate that our approach works well across datasets of different sizes, from smaller datasets such as STS-B ($\approx 5.7k$ training examples) – to the largest one – SNLI ($\approx 550k$ training examples).

5 Analysis

Impact of number of layers transferred We observed the impact of transferring a variable number of layers from unsupervised pre-training to the supervised target task. Figure 2(left) illustrates the performance of our approach on MultiNLI and RACE as a function of the number of layers transferred. We observe the standard result that transferring embeddings improves performance and that each transformer layer provides further benefits up to 9% for full transfer on MultiNLI. This indicates that each layer in the pre-trained model contains useful functionality for solving target tasks.

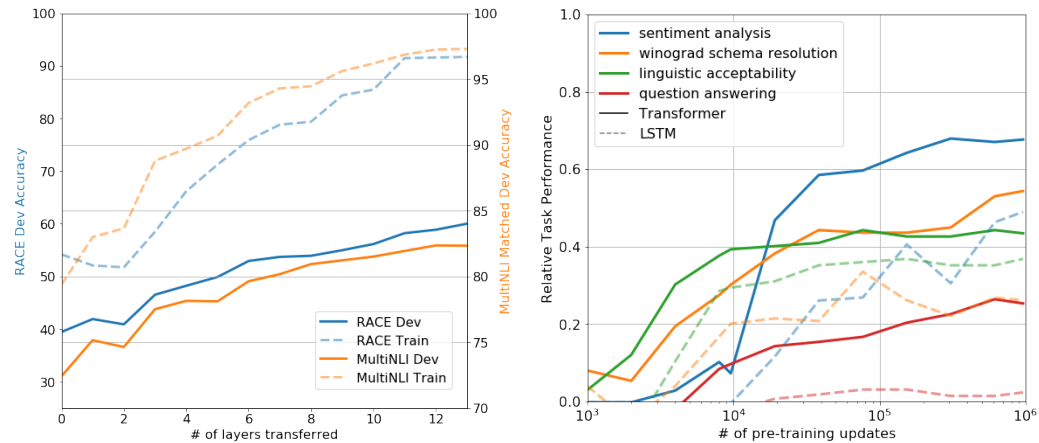


Figure 2: **(left)** Effect of transferring increasing number of layers from the pre-trained language model on RACE and MultiNLI. **(right)** Plot showing the evolution of zero-shot performance on different tasks as a function of LM pre-training updates. Performance per task is normalized between a random guess baseline and the current state-of-the-art with a single model.

Zero-shot Behaviors We’d like to better understand why language model pre-training of transformers is effective. A hypothesis is that the underlying generative model learns to perform many of the tasks we evaluate on in order to improve its language modeling capability and that the more structured

表4：语义相似度与分类结果，对比了我们的模型与当前最先进的算法。该表中的所有任务评估均基于GLUE基准测试完成。*mc*=马修斯相关系数、*acc*=准确率、*pc*=皮尔逊相关系数

方法	分类		语义相似度			GLUE
	CoLA (mc)	SST2 (召回率)	MRPC (F1)	STSB (pc)	QQP (F1)	
稀疏字节多层LSTM模型 [16]	-	93.2	-	-	-	-
TF-KLD散度 [23]	-	-	86.0	-	-	-
中南大学（混合集成方法） [60]	-	-	-	81.0	-	-
单任务BiLSTM模型 + ELMo模型 + 注意力机制 [64]	35.0	90.2	80.2	55.5	66.1	64.8
多任务BiLSTM模型 + ELMo模型 + 注意力机制 [64]	18.9	91.6	83.5	72.8	63.3	68.9
我们团队开发的微调后的Transformer语言模型	45.4	91.3	82.3	82.0	70.3	72.8

总体而言，我们的方法在所评估的12个数据集中有9个取得了新的最先进成果，在很多情况下都优于各种集成方法。我们的实验结果还表明，无论数据集规模大小如何，该方法都能保持良好性能——从像STS-B这样仅有 $\approx 5.7k$ 个训练样本的小型数据集，到规模最大的SNLI数据集（拥有 $\approx 550k$ 个训练样本），它都能稳定发挥出色作用。

5 Analysis

转移层数量的影响我们研究了从无监督预训练模型中转移不同数量的层到有监督目标任务时所产生的影响。图2（左侧）展示了随着转移层数量的变化，我们的方法在MultiNLI测验和RACE测验上的表现情况。研究结果符合常规规律：转移嵌入特征能够提升性能，而每添加一个Transformer架构层还能进一步带来收益，在对MultiNLI测验的所有层进行转移的情况下，这一提升幅度可达9%。这说明预训练模型中的每一层都具备解决目标任务所需的有效功能。

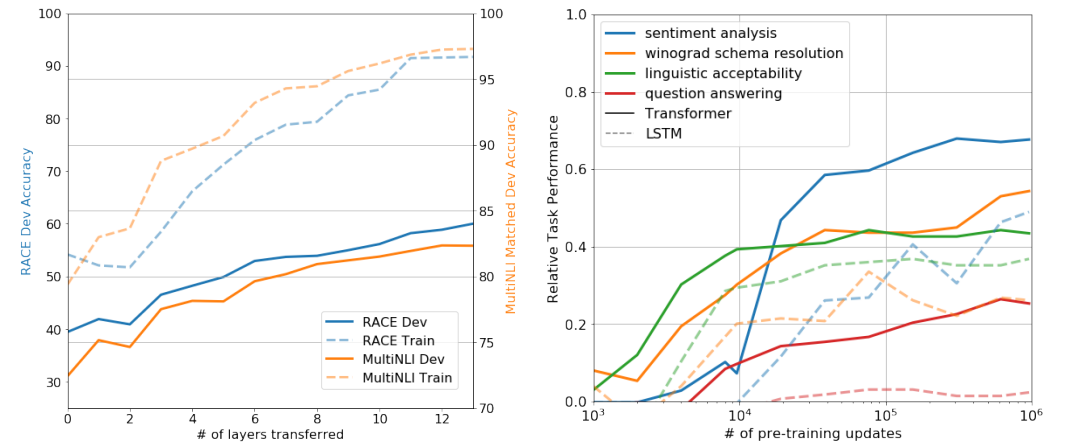


图2：（左侧）展示从预训练的语言模型中逐步转移更多层对RACE测验和MultiNLI测验的影响。（右侧）该图表呈现了随着语言模型预训练次数的增加，不同任务上的零样本性能变化趋势。各项任务的性能均以随机猜测的基准值与仅使用单个模型所能达到的当前最佳性能作为对比标准进行标准化处理。

零样本行为我们希望更深入地理解为何对Transformer架构进行语言模型预训练会如此有效。一种假设是，底层的生成式模型会为了提升自身的语言建模能力而学会完成我们所评估的诸多任务，而且这种结构更为完善的

Table 5: Analysis of various model ablations on different tasks. Avg. score is a unweighted average of all the results. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

Method	Avg. Score	CoLA (mc)	SST2 (acc)	MRPC (F1)	STSB (pc)	QQP (F1)	MNLI (acc)	QNLI (acc)	RTE (acc)
Transformer w/ aux LM (full)	74.7	45.4	91.3	82.3	82.0	70.3	81.8	88.1	56.0
Transformer w/o pre-training	59.9	18.9	84.0	79.4	30.9	65.5	75.7	71.2	53.8
Transformer w/o aux LM	75.0	47.9	92.0	84.9	83.2	69.8	81.1	86.9	54.4
LSTM w/ aux LM	69.1	30.3	90.5	83.2	71.8	68.1	73.7	81.1	54.6

attentional memory of the transformer assists in transfer compared to LSTMs. We designed a series of heuristic solutions that use the underlying generative model to perform tasks without supervised finetuning. We visualize the effectiveness of these heuristic solutions over the course of generative pre-training in Fig 2(right). We observe the performance of these heuristics is stable and steadily increases over training suggesting that generative pretraining supports the learning of a wide variety of task relevant functionality. We also observe the LSTM exhibits higher variance in its zero-shot performance suggesting that the inductive bias of the Transformer architecture assists in transfer.

For CoLA (linguistic acceptability), examples are scored as the average token log-probability the generative model assigns and predictions are made by thresholding. For SST-2 (sentiment analysis), we append the token *very* to each example and restrict the language model’s output distribution to only the words *positive* and *negative* and guess the token it assigns higher probability to as the prediction. For RACE (question answering), we pick the answer the generative model assigns the highest average token log-probability when conditioned on the document and question. For DPRD [46] (winograd schemas), we replace the definite pronoun with the two possible referrents and predict the resolution that the generative model assigns higher average token log-probability to the rest of the sequence after the substitution.

Ablation studies We perform three different ablation studies (Table 5). First, we examine the performance of our method without the auxiliary LM objective during fine-tuning. We observe that the auxiliary objective helps on the NLI tasks and QQP. Overall, the trend suggests that larger datasets benefit from the auxiliary objective but smaller datasets do not. Second, we analyze the effect of the Transformer by comparing it with a single layer 2048 unit LSTM using the same framework. We observe a 5.6 average score drop when using the LSTM instead of the Transformer. The LSTM only outperforms the Transformer on one dataset – MRPC. Finally, we also compare with our transformer architecture directly trained on supervised target tasks, without pre-training. We observe that the lack of pre-training hurts performance across all the tasks, resulting in a 14.8% decrease compared to our full model.

6 Conclusion

We introduced a framework for achieving strong natural language understanding with a single task-agnostic model through generative pre-training and discriminative fine-tuning. By pre-training on a diverse corpus with long stretches of contiguous text our model acquires significant world knowledge and ability to process long-range dependencies which are then successfully transferred to solving discriminative tasks such as question answering, semantic similarity assessment, entailment determination, and text classification, improving the state of the art on 9 of the 12 datasets we study. Using unsupervised (pre-)training to boost performance on discriminative tasks has long been an important goal of Machine Learning research. Our work suggests that achieving significant performance gains is indeed possible, and offers hints as to what models (Transformers) and data sets (text with long range dependencies) work best with this approach. We hope that this will help enable new research into unsupervised learning, for both natural language understanding and other domains, further improving our understanding of how and when unsupervised learning works.

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表5：不同任务上各类模型消融实验的分析结果。平均分是所有测试结果的非加权平均值。*mc*= 马修斯相关系数、*acc*=准确率、*pc*=皮尔逊相关系数

方法	平均分	CoLA (mc)	SST2 (召回率)	MRPC (F1)	STSB (pc)	QQP (F1)	MNLI (召回率)	QNLI (召回率)	RTE (召回率)
带辅助语言模型的完整Transformer模型	74.7	45.4	91.3	82.3	82.0	70.3	81.8	88.1	56.0
未经过预训练的Transformer模型	59.9	18.9	84.0	79.4	30.9	65.5	75.7	71.2	53.8
不带辅助语言模型的Transformer模型	75.0	47.9	92.0	84.9	83.2	69.8	81.1	86.9	54.4
带辅助语言模型的LSTM模型	69.1	30.3	90.5	83.2	71.8	68.1	73.7	81.1	54.6

Transformer架构的注意力记忆机制相较于LSTM更有助于知识迁移。我们设计了一系列启发式解决方案，利用底层生成式模型在无需进行有监督微调的情况下完成各类任务。我们在图2（右侧）中展示了这些启发式解决方案在生成式预训练过程中的效果变化。实验结果表明，这类启发式方法的性能十分稳定，并且在训练过程中持续提升，这说明生成式预训练有助于模型学习多种与任务相关的功能。同时我们也观察到，LSTM在零样本任务中的性能波动更大，这表明Transformer架构的归纳偏置更有助于知识迁移。

在处理CoLA任务（语言可接受性）时，示例的得分是根据生成模型为这些示例分配的词元对数概率的平均值来确定的，随后通过阈值判定得出预测结果；在处理SST-2任务（情感分析）时，我们会为每个示例添加样式为*very*的词元，并将语言模型的输出范围限制在*positive*和*negative*这两个词之间，进而推测模型给出更高概率的词元作为预测结果；在处理RACE测验（问答任务）时，则会在结合文档与问题作为条件的前提下，选取生成模型为它们分配的词元对数概率平均值最高的答案；而在处理DPRD [46]（Winograd框架）任务时，我们会用两个可能的指代对象替换定指代词，然后预测生成模型在替换后对序列其余部分分配的词元对数概率更高的那个指代对象。

消融实验 我们开展了三项不同的消融实验（见表5）。首先，我们考察了在微调过程中不使用辅助语言模型目标函数时该方法的性能表现。实验结果表明，辅助目标函数对自然语言推理任务和QQP任务有提升作用。总体来看，数据集规模越大，辅助目标函数带来的收益越明显，而数据集规模较小时则效果不佳。其次，我们通过在同一框架下将Transformer架构与单层2048单元LSTM模型进行对比，分析了Transformer的效果。结果显示，使用LSTM模型而非Transformer模型时，平均分数会下降5.6分。仅在MRPC这一数据集上，LSTM模型的表现优于Transformer模型。最后，我们还将其与我们直接在带监督的目标任务上训练的Transformer架构进行了比较，发现缺乏预训练会损害所有任务上的性能，使得性能相比完整模型下降了14.8%。

6 结论

我们提出了一种框架，通过生成式预训练与基于判别学习的微调，仅借助一个与特定任务无关的模型即可实现强大的自然语言理解能力。通过在包含大量连续文本的多样化语料库上进行预训练，我们的模型能够获取丰富的世界知识以及处理长距离依赖关系的能力，而这些能力随后被成功迁移用于解决问答任务、语义相似度评估、蕴含关系判定和文本分类等基于判别学习的任务，使得我们在所研究的12个数据集中有9个的数据集的性能超过了现有最佳水平。长期以来，利用无监督（预）训练来提升基于判别学习的任务性能一直是机器学习研究的重要目标。我们的工作表明，确实可以实现显著的性能提升，并且还指出了采用这种方法时哪些模型（Transformer架构）和数据集（具有长距离依赖关系的文本）效果最佳。我们希望这能为自然语言理解及其他领域中的无监督学习研究开辟新的方向，进一步帮助我们了解无监督学习的工作原理及其适用场景。

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